



CORRADO LOCATI

Senior R&D Scientist and Data Analyst

WORK EXPERIENCE

Data Analyst

London School of Economics and Political Science (Online)

08/07/2024 – 24/02/2025

Data analysis for a company delivering financial data solutions to hedge funds. The price movement of three popular stocks was compared, based on macroeconomic factors, sentiment analysis from journal articles, and surprise factor (difference between stock price and stick price prediction after each quarterly earnings report date). The entire analysis was performed with Python, and results were presented with PowerPoint.

Data analysis on data from a game shop chain with Python and R. Investigation on customer demographics, loyalty points distribution and social data trends. Linear regression and multiple linear regressions were used, together with histograms, boxplots, correlation plot, decision tree analysis, k-means clustering, word cloud, polarity, and sentiment analysis. Conclusive suggestions on which specific customer demographic (defined by gender, income, and spending score) and which parameter to monitor was given.

Diagnostic Analysis on NHS data using Python. Data on NHS appointment service were studied. Seasonal trends were found, together with causes for not going on the visit. Conclusive suggestions were given.

Data analysis for a company dealing with statistics about football players using Tableau and Excel. Players' details such as best skills, ball control, and penalties success were put in correlation with their age.

Data analyst for a supermarket chain using Excel and Tableau. Research on customers' demographics and spending habits, with suggestions on how to increase sales.

Senior Electrochemist

Elestor B.V. (Arnhem, NL)

01/01/2023 – 01/09/2024

Leading a team of 5 in researching novel platinum-based catalysts for bromine redox flow batteries.

Coordinating synthesis and characterization efforts with internal and external partners.

Conducted electrochemical tests (CV, EIS, charge-discharge) on a three-electrode setup as well as on test stands (2 and three electrodes, with 50 μm Pt wire as pseudo reference electrode).

Contributing to discussions and data analysis about corrosion technology and cell performance within other company departments. Writing SOPs and other documentation on procedures and tests.

Application Scientist

Metrohm Autolab (Utrecht, NL)

01/11/2012 – 01/11/2022

Performed laboratory experiments to write application notes covering different electrochemical domains, like corrosion (following ASTM, NACE, and ISO standards), battery testing (EIS, charge-

PROFILE

I am an Italian scientist and engineer with extensive expertise in electrochemistry, materials science, and data analysis.

I excel as an independent worker, as well as being part of international teams.

I am known for my structured approach, critical thinking, strong work ethic, and enthusiasm for new challenges.

I am open to working as a data analyst and/or senior R&D scientist. I currently live in the Netherlands, but I am open for relocation overseas.

CONTACTS

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Portfolio:

<https://corradolocati.github.io/corrado-portfolio/>

discharge, GITT, PITT), solar cells (I-V tests, IMVS, IMPS), electrocatalysis (EIS, CV with ECSA calculations), and many others.

Delivered standard operating procedures and QA/QC documentation, following ISO 9001 standard.

Actively contributed to the ASTM G01.11 committee, assisting in developing an ASTM standard for in-situ EIS measurements.

Managed product development for the battery app and the duo coin-cell holder.

Managed the chemical laboratory and demo warehouse.

Delivered global courses, mainly in introduction to electrochemistry, EIS, and how a pgstat works, and workshops for researchers and students.

Postdoctoral Researcher

TU Delft (Delft, NL)

01/01/2012 – 01/06/2012

Electrospray technology for drug delivery.

Postdoctoral Researcher

TU Delft, Kema, ARN, RUPS (Delft, NL)

01/01/2011 – 01/01/2012

Development of a membrane for Li-ion recovery from battery waste.

Postdoctoral Researcher

TU Delft, DSM (Delft, NL)

01/01/2010 – 01/07/2010

COMSOL calculations on thermal and electrical conductivities in polymer blends for materials used in LED lighting.

EDUCATION

Career Accelerator in Data Analysis

London School of Economics and Political Science (Online),

08/07/2024 – 24/02/2025

Comprehensive training in data wrangling, statistical analysis, predictive analysis, A/B testing, decision tree analysis, wordcloud, using Excel, Python, R, SQL, and Tableau. Hands-on collaboration with industry partners.

PhD in Chemical Engineering

TU Delft (Delft, NL)

01/01/2006 – 03/02/2012

Thesis: Interaction between High-Voltage Cathode Materials and Ionic Liquids for Novel Li-Ion Batteries.

Focus: Synthesis, structural characterization (XRD), and electrochemical characterization of high-voltage nanostructured cathode nanomaterials. Study in interface between the cathode material and commercial electrolytes, as well as ionic liquids. building of coin cell and pouch cell batteries were built, and tested with mainly CV, charge/discharge, and EIS. In-depth research on combining electrochemical techniques with X-ray absorption spectroscopy (EXAFS, XANES).

Second-Level Master in Material Science for Micro and Nanotechnologies

IUSS (Pavia, IT)

01/02/2005 – 01/02/2006

Synthesis, structural characterization (XRD), and electrochemical characterization (CV, charge/discharge, EIS) of high-voltage nanostructured cathode nanomaterials.

Master's Degree in Material Science
Milano-Bicocca University (Milan, IT)
01/10/1995 – 09/03/2004

Acquired both theoretical and practical knowledge in semiconductors, polymers, synthesis and characterization of materials.

Thesis: Interaction of Boron Trifluoride with Defective Sites of Binary Oxides (theoretical chemistry).

SKILLS

Data Analysis Python (numpy, pandas, matplotlib, seaborn, statsmodels, sklearn, math, scipy, wordcloud, nltk, textblob, collections), SQL, R (tidyverse, dplyr, skimr, DataExplorer, ggplot2, psych), Octave, Tableau, Microsoft Office (Word, Excel, PowerPoint).

Material science. Electrochemical Techniques like CV, EIS, GITT, PITT, LP, PDP and many others. Extensive use of Potentiostats/Galvanostats Metrohm Autolab and Biologic. Structural characterization of materials with XRD, XAS. Thermodynamic techniques (DSC, TGA), and others.

Product development & Management, Project Management Tools (Miro, Asana).

Teamwork, Problem Solving, Communication, Adaptability, Planning, Self-Management, Observation, Decision Making, Collaboration, Diplomacy, Empathy, Professionalism.

CERTIFICATIONS AND TRAININGS

Video Production for Marketing & Training – Metrohm Autolab, Utrecht, NL (12/2021)

Product Management Bootcamp – Tarigo, UK (03/2021)

Voltammetry Basics – Metrohm AG, Herisau, CH (02/2018)

Teaching Course – Metrohm AG, Herisau, CH (07/2016)

Business Presentation Skills – Van Kessel Trainings, Loosdrecht, NL (01/2015)

Spectroelectrochemistry – IFW Leibniz Dresden, De (08/2013)

Electrochemistry Winter School – University of Bath, UK (01/2013)

Scientific Writing in English PROM4 – TU Delft, NL (05/2009)

SCIENTIFIC PUBLICATIONS

Nanostructured Fe₂O₃ and CuO Composite Electrodes for Li-ion Batteries, Journal of Power Sources, 2011, 196, 6425–6432

Nanosized High-Voltage Cathode Material: Structural, Electrochemical and In-Situ Investigation, Journal of Power Sources, 2009, 189, 179–184

Interaction Between Carbon Dioxide and Ionic Liquids: Novel Electrolytes for Li-Ion Batteries, Journal of Power Sources, 2009, 189, 454–457

Mg-Doped LiNi_{0.5}Mn_{1.5}O₄ Spinel for Cathode Materials, Journal of Power Sources, 2007, 174, 847–851

Nanopowders of Spinel-Type Electrode Materials for Li-Ion Batteries, Solid State Ionics, 2006, 177, 3023–3029

Probing the Basicity of Regular and Defect Sites of Alkaline Earth Metal Oxide Surfaces, ChemPhysChem, 2004, 5, 642–651